Approximating the Score Function using Optimal Transport

Curves and Surfaces 2026

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Laboratoire de Mathématiques d'Orsay

June 11th

⏳ (Badly and Slowly) Approximating the Score Function using Optimal Transport

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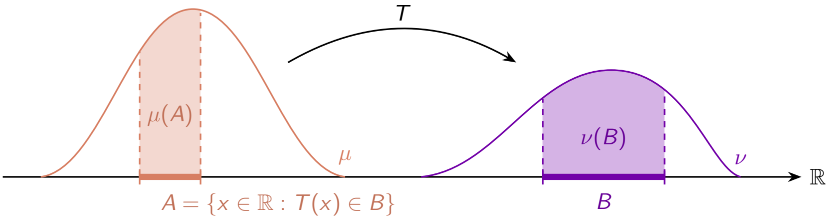
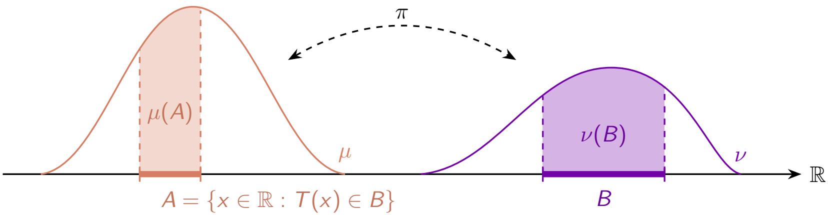
Quick reminders on Optimal Transport 1/3



- Given ground cost $c(x, y)$
- Among all maps T satisfying $\forall B,\ \nu(B) = \mu(T^{-1}(B))$
- Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x))\mu(\mathrm{d}x)$$

Quick reminders on Optimal Transport 1/3

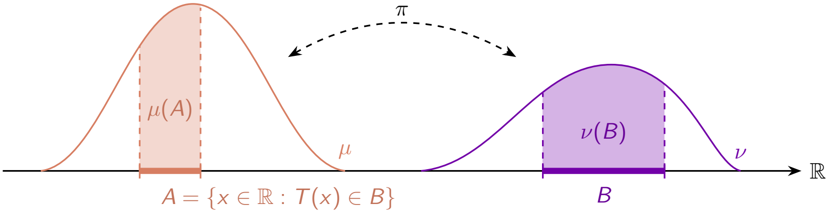


- Given ground cost $c(x, y)$
- ~~Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$~~

  Among all **transport plans** π with marginals μ and ν
- ~~Find the one minimising the average transport cost~~

$$\cancel{\int_{\mathbb{R}^d} c(x, T(x))\mu(\mathrm{d}x)} \longrightarrow \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y)\, \pi(\mathrm{d}x, \mathrm{d}y)$$

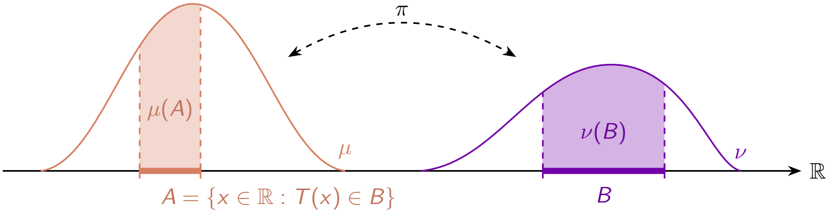
Quick reminders on Optimal Transport 2/3



- p-Wasserstein distance:

$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu,\nu)} \int d(x, y)^p \gamma(\mathrm{d}x, \mathrm{d}y)$$

Quick reminders on Optimal Transport 2/3



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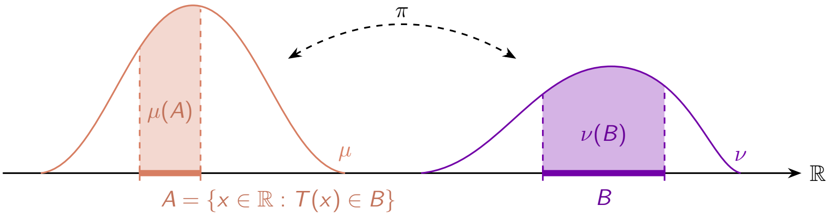
- Kantorovich duality

$$\inf_{\pi \in \Pi(\mu,\nu)} \int c(x,y)\gamma(\mathrm{d}x,\mathrm{d}y) = \sup_{\phi} \int \phi(x)\mu(\mathrm{d}x) + \int \phi^c(y)\nu(\mathrm{d}y),$$

where $\phi^c(y) := \inf_x c(x,y) - \phi(x)$.

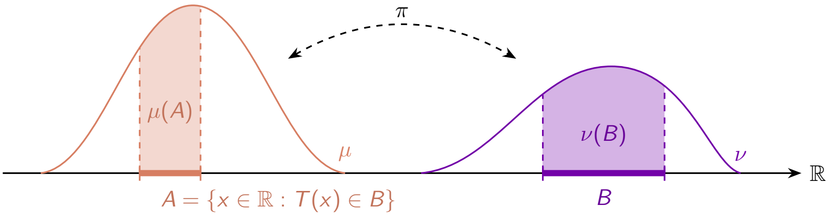
Quick reminders on Optimal Transport 3/3



- Kantorovich duality

$$\inf_{T : T_{\#}\mu = \nu} \int_{\mathbb{R}^d} c(x, T(x))\mu(\mathrm{d}x) = \sup_{\phi} \int \phi(x)\mu(\mathrm{d}x) + \int \phi^c(y)\nu(\mathrm{d}y)$$

Quick reminders on Optimal Transport 3/3



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- If an optimal transport map $T^\star$ exists, there also exists an optimal Kantorovich potential $\phi^\star$ and

$$y = T^\star(x) = \nabla\left(\frac{\|\cdot\|^2}{2} - \phi^\star\right)(x) = x - \nabla\phi^\star(x),$$

hence $-\nabla\phi^\star(x) = T^\star(x) - x = y - x$ is the **displacement** of the particle x.

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$\mathrm{d}\vec{X}_t = -\vec{X}_t \mathrm{d}t + \sqrt{2}\mathrm{d}\omega_t, \quad \vec{X}_0 \sim \mu, \quad t \in [0, T].$$

Let $\vec{\rho}_t := \mathrm{Law}(\vec{X}_t)$ and $\gamma \propto e^{-\|\cdot\|^2/2}$. Then $\vec{\rho}_t$ solves the Fokker–Planck equation

$$\partial_t \vec{\rho}_t(x) = \Delta \vec{\rho}_t(x) + \mathrm{div}(\vec{\rho}_t(x)x).$$

whose stationary solution is γ. The process $(\vec{X}_t)_t$ is a **diffusive** process.

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- **Backward process:** $\overleftarrow{X}_s := \vec{X}_{T-s}$ for $s \in [0, T]$. If $(\vec{X}_t)_{t\in[0,T]}$ is an OU process, then its time reversal $(\overleftarrow{X}_s)_{s\in[0,T]}$ solves

$$\mathrm{d}\overleftarrow{X}_s = \Big[\overleftarrow{X}_s + 2\underbrace{\nabla \ln \vec{\rho}_{T-s}(\overleftarrow{X}_s)}_{\textbf{The score function}}\Big]\mathrm{d}s + \sqrt{2}\mathrm{d}\omega_s, \quad \overleftarrow{X}_0 \sim \vec{\rho}_T.$$

Using OT to learn the score

- The Fokker–Planck equation is the 2-Wasserstein Gradient Flow (WGF) of the relative entropy $\mathcal{H}(\rho|\gamma) := \int V\rho + \int \rho \ln \rho$. That is:

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \mathrm{div}(\rho_t(x)x) = -\nabla \cdot (\rho_t(x) v_t(x)),$$

where $v_t(x) = -x - \nabla \ln \rho_t(x)$.

Using OT to learn the score

We can discretize in time the gradient flow using implicit Euler scheme (or **JKO scheme**):

$$\rho_{n+1}^{\tau} := \operatorname*{arg\,min}_{\rho \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2(\rho, \rho_n^{\tau})^2 \right\} \tag{1}$$

[1] F. Santambrogio. *Optimal Transport for Applied Mathematicians*. Vol. 87. Progress in Nonlinear Differential Equations and Their Applications. Springer, Oct. 2015. ISBN: 978-3-319-20827-5.

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As explained by Santambrogio in his book[1], we have

$$\nabla \psi_n^{c_\tau} = v_{n\tau} = -\nabla(\ln \rho_{n+1}^{\tau} + V),$$

where $\psi_n^{c_\tau}$ is a Kantorovich potential related to the transport from ρ_n^{τ} to ρ_{n+1}^{τ}. Therefore approximating the score amounts to approximating $\nabla \psi_n^{c_\tau}$.

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[1] F. Santambrogio. *Optimal Transport for Applied Mathematicians*. Vol. 87. Progress in Nonlinear Differential Equations and Their Applications. Springer, Oct. 2015. ISBN: 978-3-319-20827-5.

Stochastic Gradient Descent

Theorem 1

If $\alpha_k = \frac{R}{B\sqrt{k}}$, then

$$\forall T \geq 1, \quad \mathbb{E}\left[G_n^\tau(\bar{\theta}_T^n) - G_n^\tau(\theta_n^\tau)\right] \leq BR\left(\frac{2}{T} + \frac{1}{\sqrt{T}}\right),$$

where $\bar{\theta}_T^n$ denotes the arithmetic mean of the family $(\theta^{(k)})_{k=1}^T$ and B depends only on R and $\sup_x \|\phi(x)\|$.

Some animations

Click here to play video

Thank you for your attention!